

Name : Muhammad Owais
Reg No : FA24-BSE-054
Instructor : Sir Ehaz Mustafa
Lab #6:

Priority queue :

```
public class PriQueStatic {
    Que q0;
    Que q1;
    Que q2;
    public PriQueStatic() {
        q0 = new Que(10);
        q1 = new Que(10);
        q2 = new Que(10);}

    public void enqueue(int data, int pri) {
        if (pri == 0)
            q0.enqueue(data);
        else if (pri == 1)
            q1.enqueue(data);
        else
            q2.enqueue(data); // Priority 2 or any other value}

    public int dequeue() {
        if (!q0.isEmpty())
            return q0.dequeue();
        else if (!q1.isEmpty())
            return q1.dequeue();
        else if (!q2.isEmpty())
            return q2.dequeue();
        else {
            System.out.println("Priority Queue is Empty");
            return -1;}}

    public static void main(String[] args) {
        PriQueStatic obj = new PriQueStatic();
        obj.enqueue(20, 1);
        obj.enqueue(15, 0);
        obj.enqueue(30, 2);
        obj.enqueue(25, 3); // Goes to q2 because priority is not 0 or 1
        System.out.println("Dequeue = " + obj.dequeue());
        System.out.println("Dequeue = " + obj.dequeue());
        System.out.println("Dequeue = " + obj.dequeue());
        System.out.println("Dequeue = " + obj.dequeue());}}

    class Que {
        int[] arr;
        int tail;
        public Que(int size) {
            arr = new int[size];
```

```
tail = -1;}
```

```
public void enqueue(int data) {  
    if (isFull()) {  
        System.out.println("Queue is Full");  
    } else {  
        arr[++tail] = data;}}
```

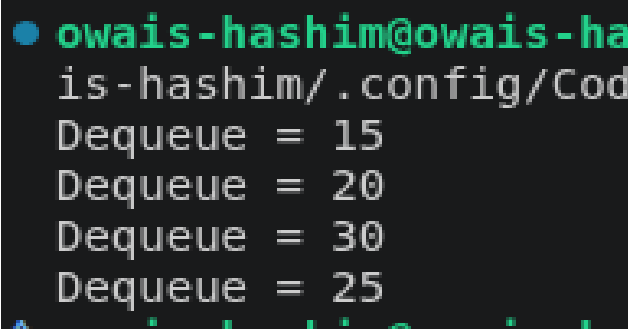
```
public int dequeue() {  
    if (isEmpty()) {  
        System.out.println("Queue is Empty");  
        return -1;  
    } else {  
        int x = arr[0];  
        tail--;  
        move();  
        return x;}}
```

```
public void move() {  
    for (int i = 0; i <= tail; i++) {  
        arr[i] = arr[i + 1];}}
```

```
public boolean isEmpty() {  
    return tail == -1;}
```

```
public boolean isFull() {  
    return tail == arr.length - 1;}}
```

Output :



```
● owais-hashim@owais-ha  
is-hashim/.config/Cod  
Dequeue = 15  
Dequeue = 20  
Dequeue = 30  
Dequeue = 25
```